

Circuit Failure Prediction and Its Application to Transistor Aging

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Abstract

Circuit failure prediction predicts the occurrence of a circuit failure before errors actually appear in system data and states. This is in contrast to classical error detection where a failure is detected after errors appear in system data and states. Circuit failure prediction is performed during system operation by analyzing the data collected by sensors inserted at various locations inside a chip. We demonstrate this concept of circuit failure prediction for a dominant PMOS aging mechanism induced by Negative Bias Temperature Instability (NBTI). NBTI-induced PMOS aging slows down PMOS transistors over time. As a result, the speed of a chip can significantly degrade over time and can result in delay faults. The traditional practice is to incorporate worst-case speed margins to prevent delay faults during system operation due to NBTI aging. A new sensor design integrated inside a flip-flop enables efficient circuit failure prediction at a low cost. Simulation results using 90nm and 65nm technologies demonstrate that this technique can significantly improve system performance by enabling close to best-case design instead of traditional worst-case design.

1. Introduction

The idea behind *circuit failure prediction* is to predict the occurrence of a failure during normal system operation **before** the appearance of any error that can result in corrupt system data and states. This is in contrast to classical *error detection* where a failure is detected **after** errors appear in system data and states. Failure prediction can be performed in multiple ways – the basic principle is to insert a wide variety of “sensors” at various locations inside a chip. During normal system operation, these sensors collect information about various system parameters over time. Examples of such system parameters range from temperature, voltage, ring oscillator delays to complex relative timing relationships among logic signals. The data collected by the sensors is analyzed on-chip or off-chip to identify “anomalies” and predict failures. Sensor designs and data analysis techniques can widely vary depending on failure sources and their sensitivities. The purpose of this paper is to introduce the concept of circuit failure prediction and apply this concept to a dominant reliability mechanism – PMOS transistor aging induced by Negative Bias Temperature Instability (NBTI).

Table 1.1 compares and contrasts various aspects of failure prediction and error detection. Unlike error detection, a major limitation of failure prediction is that all circuit failures cannot be predicted, e.g., radiation-induced soft errors. However, failure prediction is suitable for failures caused by major reliability mechanisms such as NBTI-induced transistor aging because of the gradual nature of degradation. NBTI is expected to be dominant in future scaled-CMOS technologies [Borkar 05, Massey 04]. Failure prediction enables a system to initiate corrective measures before system data and states actually get corrupted by errors – this prevents error propagation and error detection latency problems associated with classical error detection. Circuit failure prediction can be performed at a low

cost since it targets gradual failures and relies on analysis of data collected over several clock cycles. Such data analysis can also be extremely beneficial for self-diagnostics.

The major contributions of this paper are:

- Introduction of the concept of circuit failure prediction, and demonstration of its practicality and effectiveness for NBTI-induced PMOS transistor aging.
- Design of special sensors for predicting aging and demonstration of their efficiency and effectiveness.
- Demonstration of new design techniques that enable close to best-case performance unlike traditional design techniques that impose worst-case speed margins.

Table 1.1 Circuit failure prediction vs. error detection.

Circuit failure prediction	Error detection
Before errors appear	After errors appear
No corrupt data or states	Corrupt data & states
Can collect data over several cycles before predicting failure	Long error detection latency causes data integrity problems.
Generally inexpensive	Generally expensive
Self-diagnostics possible	Limited self-diagnostics
Incorrect prediction can be a problem	Insufficient coverage can be a problem
All failures not predictable	General applicability
Both can be efficiently combined	

Section 2 introduces NBTI-induced PMOS transistor aging and its impact on future designs. Section 3 explains how the concept of failure prediction can be used to overcome transistor aging challenges. New sensor designs for predicting aging failures are presented in Sec. 4. Simulation results demonstrating the effectiveness and efficiency of such designs are presented in Sec. 5. Section 6 discusses a few corrective measures that can be initiated upon aging failure prediction. Section 7 discusses related work, followed by conclusions and future work in Sec. 8.

2. NBTI-induced PMOS Transistor Aging

Transistors age over time – this can have significant impact on performance of future circuits [Borkar 05]. NBTI is a dominant aging mechanism causing PMOS threshold voltage degradation over time, resulting in degraded drive current and increased delay. The cause of PMOS NBTI is associated with the generation of the interface traps and (positive) oxide-fixed-charge by electrochemical reaction of holes and Si-H bonds at the Si-SiO₂ interface when the PMOS transistor is turned on, i.e., the gate-source is negatively biased. This problem became significant at 90nm technology node [Agostinelli 05, Chen 03, Reddy 02, Schroder 03], and is expected to get significantly

worse in future generations [Massey 04, McPherson 06, Reddy 02].

The amount of threshold voltage degradation of a PMOS transistor due to NBTI depends on several factors – the amount of time elapsed, temperature and voltage profiles experienced by the PMOS transistor, and the workload, which determines the amount of time the PMOS transistor is on – as discussed next [Paul 05, Vattikonda 06]:

$\Delta V_{th} \propto t^n$, where ΔV_{th} is the PMOS threshold voltage degradation, t is the amount of time, and $n = 0.25$ is typically used for current technologies;

$\Delta V_{th} \propto e^{-(E_a/KT)}$, where E_a is the activation energy of Si, K is the Boltzman's constant, and T is the operating temperature;

$\Delta V_{th} \propto e^{[(|V_{gs}| - |V_{tp}|)/E_0] t_{ox}}$, where t_{ox} is the gate oxide thickness, V_{gs} is the gate-source voltage and $E_0 = 2.0$ MV/cm.

The power law dependence on time results in fast degradation in the beginning of lifetime. The rate of degradation slows down over time.

Chip-level effects of NBTI can be very complex. For example, delays of different paths on the same chip or same paths on different chips can degrade very differently. This can lead to changes in timing critical paths over time. For example, a circuit path which is not timing critical right after manufacture may become timing critical later.

The current industrial practice to cope with transistor aging is to slow down the clock frequency, i.e., add timing margin, during design / test based on the worst degradation the transistors might suffer during lifetime due to worst temperature, worst voltage, worst percentage of time the PMOS transistors are on, etc. This is called *guardbanding*. However, as discussed earlier, the actual aging of a chip depends on its field usage – temperature, voltage, and workload. With worst-case guardbanding, all chips may suffer significant reduction in speed although most of them may not be stressed to worst levels in the field. Another phenomenon which makes worst-case guardbanding more difficult in future technologies is that NBTI-induced aging has a statistical component, i.e. similar nominal devices under similar environmental and workload conditions may age at different rates [McPherson 06]. Hence, worst-case guardbanding must also account for the spreads (standard deviations) of aging distributions and must guarantee a high degree of confidence (e.g., > 99.9% confidence). This further worsens worst-case guardbands. The circuit failure prediction concept, presented in this paper, enables elimination of such worst-case guardbands.

3. Circuit Failure Prediction for Transistor Aging

The overall circuit failure prediction idea for overcoming NBTI-induced PMOS aging is summarized in Fig. 3.1. Instead of introducing a worst-case guardband over the entire lifetime of a design (e.g., 7 to 10 years for enterprise systems or 3 to 5 years for PCs), the system starts off with a small worst-case timing guardband that guarantees correct circuit operation even with worst aging over a short period of time, e.g., 15 days. We refer to this timing guardband as the *guardband interval*. The choice of this period is dependent on overall circuit properties and will be discussed in details in Sec. 5. During this 15-day period of operation, various parts of a chip will age at various rates depending on the workload, voltage and temperature profiles. During this time, specially designed on-chip sensors collect lots of data about relative aging of various circuit paths. At the end of 15 days, the data collected by the sensors is analyzed to check

whether there has been enough aging that requires adjustment of the 15-day timing guardband. If yes, the design adapts itself based on its operation history by adjusting the guardband based on the estimated amount of actual aging using adaptive self-correction techniques discussed in Sec. 6. The distinct difference of this approach from error detection is that system data and states never get corrupt. The 15-day worst-case timing guardband in Fig. 3.1 ensures that the system will produce correct outputs even when the system has aged enough to require adjustment of the 15-day timing guardband at the end of 15 days. Note that, the 15-day period does not necessarily have to be constant over the entire lifetime.

In an ideal situation, the initial margin should be as short as possible instead of 15 days in Fig. 3.1. However, there are several trade-offs that influence this initial guardband:

- Design of sensor circuits requires some margin. Otherwise, sensor designs may get extremely complex;
- Some data must be collected by the sensors to identify anomalies before failures can be predicted; and,
- The rate of degradation is very high in the very beginning.

These aspects are discussed in more details in Sec. 5. There are two major components of circuit failure prediction:

1. Failure prediction sensor design, and
2. Self-correction techniques.

The major focus of this paper is on failure prediction sensor designs (described in details in Sec. 4 and 5). We discuss possible self-correction options in Sec. 6.

4. Sensor Design for Transistor Aging Failure Prediction

The success of failure prediction depends on the accuracy of data collected by on-chip sensors. For PMOS aging, use of structures such as on-chip ring oscillators [Reddy 02] and temperature sensors can be helpful but is inadequate (details in Sec. 5.6). This is because the actual amount of aging strongly depends on the workload (the percentage of time the PMOS is on). The workload of an on-chip ring oscillator can be very different from the workload experienced by circuit blocks – hence, the amount of aging can be very different.

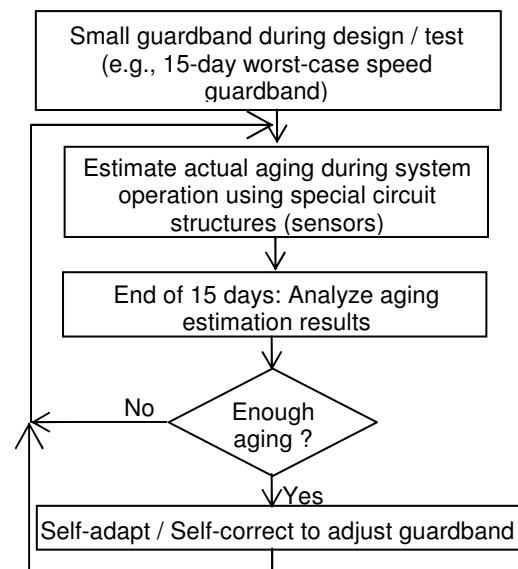


Figure 3.1. Circuit Failure prediction for NBTI-induced aging.

Another sensor design approach is to duplicate a few timing-critical paths [Tschanz 02], monitor their aging, and draw conclusions on the overall aging of a chip based on the observed aging of the duplicated critical paths. This approach suffers from two problems: Workloads experienced by duplicated paths are different from actual circuit paths, and, as discussed before, critical paths may change over time. We validated this fact through SPICE-level simulations using published NBTI-induced PMOS aging models [Paul 05].

Our sensor design approach is to modify a standard latch or flip-flop by inserting a “monitoring” circuit block which detects any ‘significant’ shifts in delay of the combinational logic whose output is connected to the data input of that latch or flip-flop. In this paper, we focus on rising-edge-triggered flip-flop-based designs – however, the presented techniques are applicable for falling-edge-triggered flip-flops and latches as well. The monitoring circuit block is based on the concept of *stability checking* during the guardband interval by detecting signal transitions during the guardband interval, also referred to as *guardband violation*. Figure 4.1 illustrates this point.

Guardband violation at the combinational logic output means that the combinational logic input stimulus exercises one or more paths that have aged enough to creep into the guardband interval and is now very close to turning into a delay fault. Note that, the flip-flop will still continue to capture correct logic values from the combinational logic – that is the basic difference from error detection. Figure 4.2 shows the block diagram of a design consisting of a flip-flop with a built-in aging sensor. An aging sensor integrated inside a flip-flop has three components: a stability checker (Fig. 4.3), a delay element to create an interval during which stability checking is performed (Fig. 4.4), and an output latch to store stability checking results. Details of each aging sensor component are discussed later in this section. The delay element and the output latch may be shared among multiple flip-flops for power and area savings. Depending on how scan flip-flops are designed, it may be possible to reuse the scan portion of a design for the output latch (e.g., utilizing 3-latch or 4-latch scan flip-flop designs [Das Gupta 81, Kuppaswamy 04]).

Figure 4.3 shows the design of a stability checker and Fig. 4.5 shows the corresponding waveforms. In the beginning of a clock cycle when $Clock = 1$ (i.e., $Clock_b = 0$), PMOS transistors T1 and T5 in Fig. 4.3 are on (NMOS transistors T3 and T7 are off), and the stability checker output $Out = 0$. This is called the *precharge phase* of the stability checker. The delay element (Delay) in Fig. 4.3 (and shown in detail in Fig. 4.4) introduces a delay of ‘ $T_{clk}/2 - T_g$ ’ (assuming 50% duty cycle of $Clock$). Here, T_g is the guardband interval (Fig. 4.1). This guarantees that transistors T3 and T4 (and transistors T7 and T8) are both on during the guardband interval T_g (called the *evaluate phase*). During the guardband interval, PMOS transistors T1 and T5 are turned off. Stability Checker output Out becomes 1 if and only if the combinational logic output OUT transitions from 1 to 0 or 0 to 1 once or multiple times during the guardband interval, i.e., the guardband is violated. The output latch in Fig. 4.2 is responsible for holding the state of the sensor output whenever the stability checker catches any guardband violation (i.e., stability checker output is 1). Example waveforms are shown in Fig. 4.5.

For the presented technique to work for NBTI-induced PMOS aging, it is critical that the performance of the aging sensor itself does not degrade significantly due to PMOS transistor aging. Degradation of PMOS transistors T1 and T5 is not a major concern because the precharge phase, when T1 and

T5 affect the performance of the stability checker (i.e., when Out is precharged to 0), is generally sufficiently long.

Design of an aging resistant delay element is shown in Fig. 4.4. Since NBTI-induced PMOS aging is a slow process, aging sensors can be turned off most of the time to ensure that they do not age significantly. This requires an additional slow global input signal, *Monitor* in Fig. 4.4. This signal may be derived from the scan enable signal. When $Monitor = 1$, aging monitoring is turned on and the delay element outputs a delayed version of $Clock_b$. When $Monitor = 0$, aging monitoring is turned off and the delay element produces 1. Turning aging monitoring off also reduces power consumption of the delay element. However, we must ensure that the source-gate junctions of all PMOS transistors used in generating the guardband interval must not be forward biased when aging monitoring is turned off. The aging resistant delay element design in Fig. 4.4 accomplishes this in the following way discussed next.

When $Monitor = 1$, aging monitoring is turned on and the series of NAND gates (G2, G3, G4) introduce the delay required to produce a delayed version of $Clock_b$. When $Monitor = 0$, each NAND gate (G2, G3, G4) produces a 1 at its output irrespective of the other input signal. As a result, the corresponding PMOS transistor in the following NAND gate that this output is connected to is off and doesn’t age. Note that, the PMOS transistor in a NAND gate connected to the *Monitor* signal continues to age. However, this PMOS transistor will not affect the delay of the circuit path which produces *delayed Clock_b* since it is in parallel to the other PMOS transistor which is off when $Monitor = 0$.

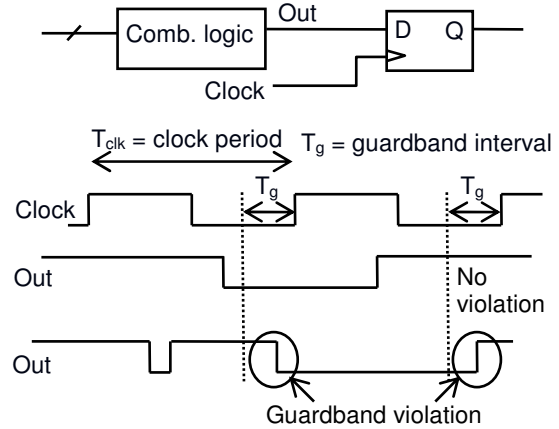


Figure 4.1. Guardband violation due to transistor aging.

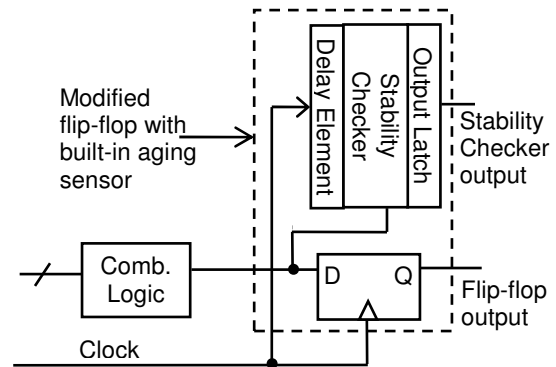


Figure 4.2 Flip-flop with built-in aging sensor.

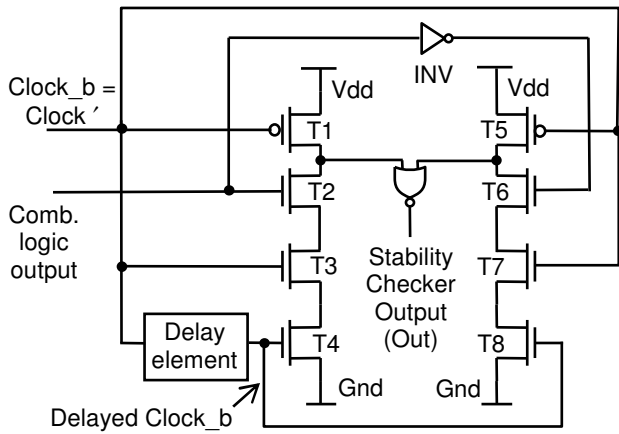


Figure 4.3: Aging resistant Stability Checker design.

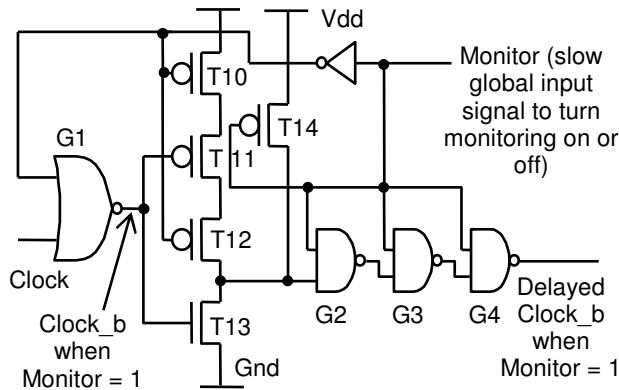


Figure 4.4: Aging resistant delay element design.

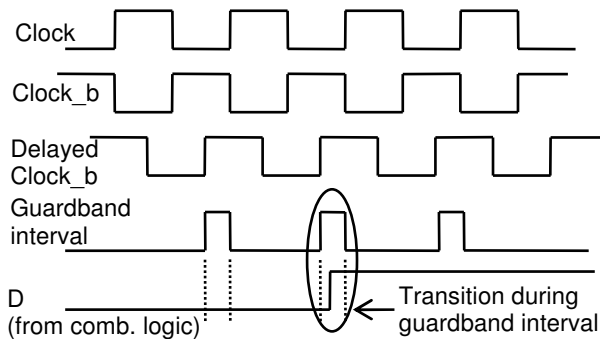


Fig. 4.5: Timing Diagram for aging sensor.

To prevent aging of the first PMOS transistor connected to the clock input of the delay element, we use a NOR gate (G1) instead of NAND. The PMOS transistor inside G1 connected to the *Clock* input is off (and does not age) when *Monitor* = 0. This NOR gate inverts *Clock* to generate *Clock_b* when *Monitor* = 1, and produces 0 when *Monitor* = 0. The NOR gate output cannot be directly connected to the input of NAND gate G2 because the corresponding PMOS transistor on the circuit path producing *delayed Clock_b* will age when *Monitor* = 0. Instead, the output of G1 is connected to an inverter formed out of T11 and T13. When *Monitor* = 0, T10 and T12, in series with T11, are turned off ensuring that T11 does not age. When *Monitor* = 0, T13 is off and T14 is on producing 1 at the input of G2. This ensures that the PMOS transistor of G2 connected to the circuit path producing *delayed Clock_b* does not age when *Monitor* = 0.

An alternative aging sensor design based on the stability

checking concept is described in Fig. 4.6. The idea is to pre-sample the combinational logic output (using a delay element) and compare it with the actual captured value using an XOR gate. Note that, ' T_g ' in Fig. 4.6 is equal to the guardband interval and not ' $T_{clk}/2 - T_g$ ' used in Fig. 4.3. If the guardband is violated due to aging, then the two samples will be different and a mismatch will be reported by the XOR. This technique is simple and the power impact of the delay element here can be smaller than the delay required in Fig. 4.2 since the activity factor of the combinational logic output is lower than the clock. However, if the aging sensors are mostly turned off and are turned on as and when required, then the power impact will be very small for design shown in Fig. 4.2 design as well. The design in Fig. 4.6 is prone to invalidation due to hazards (similar to invalidation during path delay testing). Moreover, unlike the design in Fig 4.2, the delay element cannot be shared among multiple stability checkers and contributes towards larger area and power. For both sensor designs, additional resolution can be obtained by adding additional delay elements.

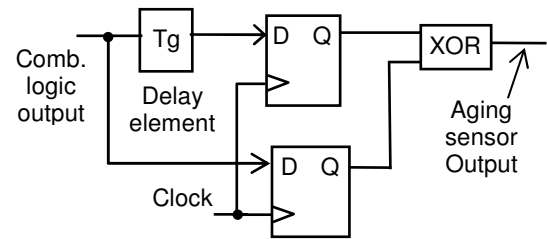


Fig. 4.6: Alternative aging sensor design.

5. Simulation Results and Analysis

Most of the results presented in this section are based on simulations performed using a 65nm technology from a leading semiconductor company (referred to as *Tech A* in this paper). In addition, we validated our designs using 90nm technologies from two other leading semiconductor companies (referred to as *Tech B* in this paper) and the publicly available 65nm Predictive Technology Model (PTM) [Zhao 06, PTM] (not reported in this paper).

5.1. Performance, Power and Area Impact

First, we report performance, power and area penalties imposed by the aging sensor design in Fig. 4.2 obtained from simulations using 65nm technology (*Tech A*). In order to quantify these characteristics, we simulated an FO4 inverter chain (activity factor of 10%) with our modified flip-flop design with a built-in aging sensor (Fig. 4.2). The simulation results are summarized in Table 5.1. The percentage power increase is computed as the percentage increase in total power of the inverter chain and master-slave flip-flop together with the built-in aging sensor (delay element, stability checker and latch) inside the flip-flop. As shown in Table 5.1, performance penalty due to additional loading on the inverter chain output is negligible (<1%). Most of the power in the sensor is dissipated due to switching in the delay element. Hence, in Table 5.1, the power impact is very small when the sensor is off. Moreover, sharing of the delay element among multiple stability checker blocks (4 in Table 5.1) saves power and also reduces the number of transistors required. A global signal (*Monitor*) needs to be routed throughout the chip to turn the aging sensor on or off.

Table 5.2 summarizes chip-level costs associated with the aging sensors. Overall chip-level power penalty is very small because the sensors do not need to be used for all flip-flops and do not need to be activated all the time. For example, producing

of a synthesized OpenRISC processor design (available from www.opencores.org) shows that roughly 40% of all flip-flops require built-in sensors for pessimistic situations in which circuit path delays can increase by up to 25% due to transistor aging. This estimate is made by identifying each flip-flop with D-input connected to at least one circuit path with slack less than 25% of the total cycle time. In that case, the chip-level power penalty of using aging sensors will be 4% when all aging sensors are on. Since, aging sensors are expected to be on for only a small fraction of time, chip-level average power penalty further reduces to 0.4% if the aging sensors are on 10% of the time (which is a pessimistic assumption). The flip-flop with built-in aging sensor can be a standard cell in the technology library. Hence, we do not expect any major impact on design and test flows.

Table 5.1: Flip-flop-level performance, power and area tradeoffs of aging sensors (chip-level penalties are lower).

Tradeoffs/ Penalties	No sharing of Delay element & latch	Delay element & latch shared among 4 blocks
Performance penalty	< 1%	< 1%
Power penalty when aging sensors are off (most of the time)	0.3%	0.1%
Power penalty when aging sensors are on (small fraction of time)	12.5%	7.5%
Area (Transistor count per sensor)	51	24
Global signal	1 (slow)	1 (slow)
Minimum guardband T _g	20ps	20ps

Table 5.2: Chip-level performance, power and area tradeoffs of aging sensors for synthesized OpenRISC core.

Speed penalty	< 1%
% flip-flops with built-in aging sensors (pessimistically assuming 25% increase in path delays due to aging)	< 40%
Average power penalty (aging sensors on 10% of the time)	< 0.4%
Additional global signal	1 (slow)
Scan path reuse	Possible
Die size increase	Not expected
Design flow impact	Minimal

5.2. Aging Sensor Charge Sharing, Leakage and Aging

Since the circuit shown in Fig 4.3 is a dynamic circuit, we must be very careful about charge sharing and leakage problems. Our Fig. 4.3 circuit does not suffer from the charge sharing problem since, during the evaluate phase, either the transistor closest to the output node (T2 or T6) is off, or all intermediate nodes in the NMOS stack are pulled down.

The stability checker circuit shown in Fig. 4.3 is pre-charged during the period when *Clock_b* is low and is susceptible to leakage when *Clock_b* is high. Simulation results using 90nm technology (*Tech B*) shows that the NOR Gate output in Fig. 4.3 does not flip even for clock frequencies as low as 2MHz (*Tech A* was not available to us during the entire time of this project). For future technologies the effect of leakage may become more significant. In that case, the problem can be

mitigated by placing PMOS transistors in parallel to T1 and T5 that are controlled by the *delayed Clock_b* signal.

We used the aging simulator of a leading semiconductor company for the 65nm technology (*Tech A*) to confirm that PMOS aging along the clock path in the delay element is extremely small due to design considerations discussed in Sec. 4.

5.3. Extended Guardband for Reliable Aging Sensor Operation

If the combinational logic output transitions very close to either the rising or the falling edge of the guardband interval formed by the *Clock_b* and *delayed Clock_b* signals, then the transition may not be reliably detected by the stability checker. This may be due to non-zero delay of the inverter INV (Fig. 4.3) or incomplete discharge of either of the NMOS stacks (Fig. 4.3), and may be treated as setup time violation of the stability checker block. To overcome this problem, we add a small margin to the aging guardband interval to create an *extended* guardband (Fig 5.1). Any transition close to the rising edge of this extended guardband will result in one of the two possibilities: 1. The transition is detected indicating violation of the guardband; or, 2. The transition is not detected because it occurred very close to the rising edge of the guardband period (within the added timing margin). This margin also ensures that logic values propagating along any combinational logic path (including the one that suffers the worst degradation) will settle down well before the rising edge of *Clock*. Hence, any signal transition during the guardband period will be reliably captured by the stability checker. Our simulation results indicate that a 20ps margin is sufficient to create the extended guardband for both *Tech A* and *Tech B*.

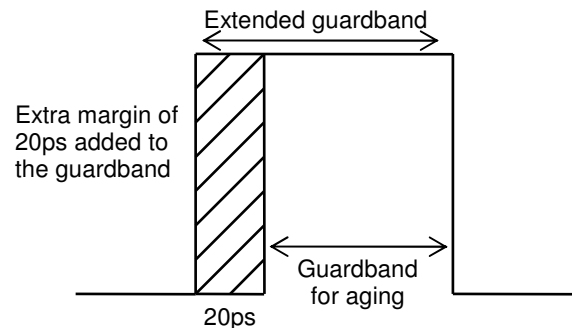


Figure 5.1. Additional 20ps setup time margin to reliably detect signal transitions on combinational logic output.

5.4. Impact of Process Variations on Aging Sensor Design

Since the aging sensor circuit uses a delay element (Fig. 4.4) to generate the guardband interval, process variations, e.g., transistor channel length and threshold voltage variations, in the delay element design can result in undesirable variations in the guardband interval. We analyze the effects of process variations on the delay element design by performing Monte-Carlo simulations using the 90nm technology (*Tech B*). For each transistor, we varied channel length and threshold voltage following Gaussian distributions with means obtained from nominal values in the technology file and 3σ values equal to 30% of the nominal values. For simulation purposes, threshold voltage variation from the nominal value is modeled as an equal change in the gate-to-source voltage, without changing the threshold voltage itself. The variations are assumed to be independent for all transistors. Any variation in the delay

through the delay element (whose nominal delay is 'Tclk/2 – Tg') will result in a variation in the length of the guardband interval during which the stability checker detects signal transitions at the combinational logic output (Fig. 4.5). Figure 5.2 shows the distribution of the variations in the guardband interval imposed by the variations in the delay element design (Fig. 4.4) obtained from 250 simulation runs. The nominal length of the guardband interval is 120ps.

There are multiple ways to cope with process variations. We discuss a few possibilities that are suited for our application based on the fact that delays through delay elements can be characterized during manufacturing test.

If the guardband interval is shorter than the nominal value (i.e., the delay through the element is greater than its nominal value), it implies that the stability checker may miss a signal transition that will otherwise be detected for nominal value of the length of the guardband interval. This problem can be resolved by adjusting the number of days that correspond to the initial guardband interval (e.g., if the nominal value of 120ps guardband corresponds to 15 days of guaranteed correct operation, then 100ps guardband means fewer than 15 days of guaranteed correct operation).

If the guardband interval is longer than the nominal value (i.e., the delay of the delay element is smaller than its nominal value), then the stability checker effectively checks for signal transitions during a window longer than the intended nominal guardband interval. This may lead to false positives (i.e., signal transitions may be detected by the stability checker even if the circuit does not age). There are two ways to overcome this problem: 1. Pessimistically design for a longer guardband interval after accounting for process variations, or, 2. Design the delay element such that its delay can be increased during manufacturing test.

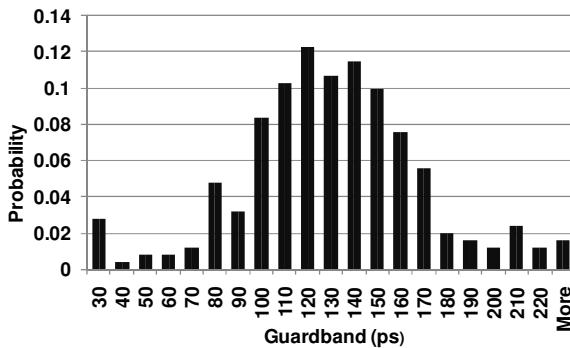


Fig 5.2: Monitoring window size under process variations.

5.5. Guardband Reduction Effectiveness using Aging Sensors

Table 5.3 presents analytical results for the amount of guardband reduction that can be obtained by using our failure prediction technique for systems with clock frequencies of 1 GHz, 2 GHz and 3 GHz. We use the flowchart of Fig. 3.1 – instead of starting with an initial 15-day guardband, we sweep the initial guardband from 1 day to 1 year. In addition, we sweep the amount of 7-year delay degradation due to transistor aging from 5% to 40% of the clock period. Here is a sample calculation to illustrate how the numbers in Table 5.3 are obtained. It follows from [Paul 05] that:

$$\frac{\Delta\tau}{\tau} = Kt^{0.25} \quad (1)$$

where τ is the gate delay, $\Delta\tau$ is the change in gate delay due to aging, t is the time in days and K is a constant proportionality.

Consider a hypothetical example of a system with 1 GHz and suppose that aging over 7-years imposes a guardband of 25% of the clock period (= 250ps). Suppose that we use circuit failure prediction instead, using our aging sensor of Fig. 4.2 for a system, starting off with a 60-day guardband (instead of a 7-year guardband).

$$\begin{aligned} \text{7 year degradation} &= 0.25 \times 1\text{ns} \\ \text{Degradation during first 60 days} &= 0.25 \times \left(\frac{7 \times 365}{60}\right)^{-0.25} \times 1\text{ns} \\ \text{\%reduction} &= \frac{7 \text{ year guardband} - (60 \text{ day guardband} + 20\text{ps})}{7 \text{ year guardband}} \times 100\% \\ &= 53\% \end{aligned}$$

Note that, we add an additional 20ps to ensure reliable operation of our aging sensor design as discussed in Sec. 5.3. It is clear from Table 5.3 that significant guardband reduction is obtained using our circuit failure prediction technique. Negative numbers in Table 5.3 correspond to the cases in which circuit failure prediction using our aging sensor design is not beneficial in reducing the guardband. In principle, a shorter initial guardband implies better performance (e.g., 1 day instead of 15 days). However, that requires very frequent monitoring and periodic on-line self-test to ensure that enough circuit paths have been exercised (which is extremely important for circuit failure prediction).

Table 5.3: Percentage reduction in guardband enabled by circuit failure prediction using the aging sensor design of Fig. 4.2.

Clock period	7 yr guardband (%clock period)	Initial guardband interval (days)			
		1	5	15	60
1 GHz	5	46	39	32	21
	10	66	59	52	41
	25	78	71	64	53
	40	81	74	67	56
2 GHz	5	6	-1	-8	-19
	10	46	39	32	21
	25	70	63	56	45
	40	76	69	62	51
3 GHz	5	-34	-41	-48	-59
	10	26	19	12	1
	25	62	55	48	37
	40	71	64	57	46

5.6. Aging Sensors vs. Temperature Sensors for Circuit Failure Prediction

An alternative to our aging sensor approach to circuit failure prediction is to use temperature sensors in various parts of a chip, and then predict the aging of a chip based on the temperature sensor readings. There are several practical issues in deploying such temperature sensors (e.g., where to insert, sensitivities to global vs. local temperature changes, etc.). We assumed perfect temperature sensors for comparison purposes. Circuit failure predictions based on idealized temperature sensors will have accurate temperature information but must assume worst-case values of the percentage on-time for PMOS transistors (e.g., 100% in the worst-case or calibrated based on

simulations of worst-case workloads) and worst-case values of voltage. On the other hand, circuit failure prediction using aging sensors has accurate delay degradation information, but requires an additional 20ps margin for reliable prediction.

In Fig. 5.5a, we show the benefits of circuit failure prediction using aging sensors (i.e., smaller guardbands) over temperature sensors over a range of temperature values. For this plot, it is pessimistically assumed that the PMOS transistors are on 50% of the time. Figure 5.5b illustrates the benefits of aging sensor-based over a range of values for the percentage of time PMOS transistors are on (0%-100%). For this plot, the operating temperature is assumed to be 55°C. Both plots in Fig. 5.5 demonstrate that aging sensors enable up to 2x reduction in guardband compared to idealized temperature sensors. Note that, temperature sensors are inaccurate in practice unlike idealized temperature sensors assumed in this analysis. This implies that better benefits are possible from circuit failure prediction using aging sensors described in this paper. The aging sensors (Fig. 4.2) are able to ensure close to best-case performance since they do not rely on worst case conditions of voltage, temperature or workload.

6. Self-Correction Options

As shown in Fig. 3.1, failure prediction alone isn't enough. We require self-correction / self-adaptation techniques. Some self-correction / self-adaptation for NBTI-induced transistor aging include:

1. Variable clock frequency: The clock frequency may be reduced over time depending on the actual usage and aging of a chip in the field;
2. Adaptive body bias: Since the primary effect of NBTI is to increase the magnitude of threshold voltage of a PMOS transistor, forward biasing the source-body junction can compensate for the change. Adaptive body bias has been shown to be effective for compensating die-to-die parameter variations in [Tschanz 02] and for reducing power in high-performance processors [Martin 02]. Body bias has also been used for leakage improvement and process compensation. The tradeoff involved in using forward body bias for aging is the decrease in magnitude of threshold voltages for PMOS transistors that have not aged. This will result in increased leakage power;
3. Adaptive power supply: Another approach is to increase the voltage supply (and hence reduce delay) whenever the stability checker circuit detects a violation. It must be ensured that the increased supply is within the Vmax limit set by the technology. Increasing the supply voltage also unnecessarily speeds up parts of the circuit that have not aged and also accelerates aging.

7. Related Work

There has been some prior work on software failure prediction such as crash prediction [Gross 03, Lin 90]. Prior work on Dynamic Reliability Management (DRM) [Simunic 04, Srinivasan 04a, 04b, 05, Urmanov 04] has similarities with the circuit failure prediction technique in this paper at a coarse-grained level. Unlike the DRM approach, our circuit failure prediction idea is applied at a very fine-grained level assisted by special circuit structures. It can be viewed as a general way of speed-binning during normal operation. In fact, DRM techniques can benefit from the sensor structures used for circuit failure-prediction. Our circuit failure prediction technique complements timing error detection techniques such as [Ernst 03, Franco 94, Nicolaidis 99]. This is because our technique enables circuit failure prediction and timing-diagnostics but is

not intended for detecting errors and recovery. Moreover, since we perform failure prediction, our self-correction options can be application-independent (Sec. 6) unlike processor pipeline recovery approaches. Stability checking was also used in [Franco 91, 94]. Unlike [Franco 91, 94], we use pre-sampling for aging failure prediction purposes rather than post-sampling. A few attempts have been made to suppress the effects of NBTI both in the process and circuit level [Kumar 05, Paul 06, Scarpa 06]. The circuit level approaches are static in nature based on estimation of worst-case reliability degradation.

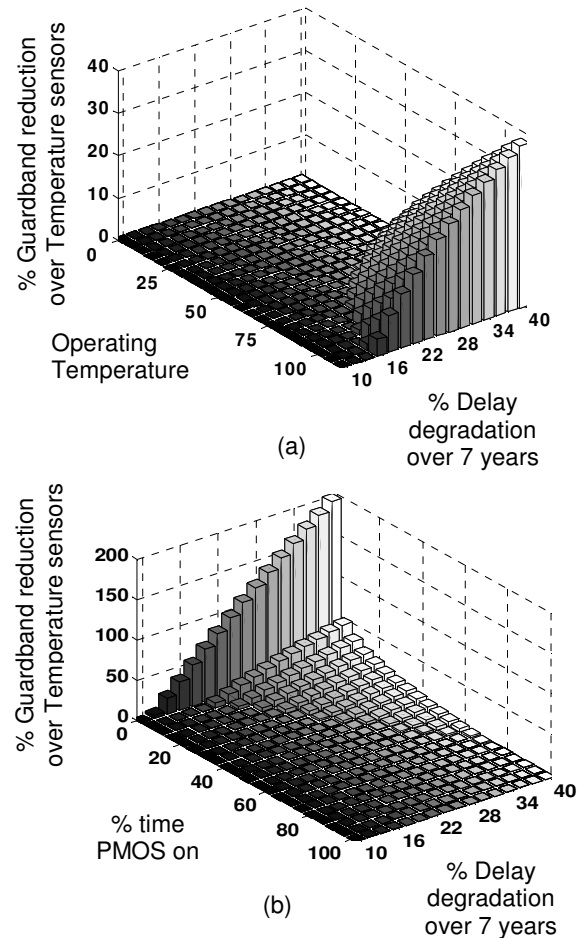


Fig. 5.5: Aging sensors vs. temperature sensors for circuit failure prediction (a) Sensitivity to operating temperatures. (It is assumed that the PMOS is on 50% of the time). (b) Sensitivity to % time PMOS is on. (It is assumed that the operating temperature is 55°C).

8. Conclusions and Future Work

Circuit failure prediction provides an effective way of overcoming reliability challenges in scaled-CMOS technologies, e.g., NBTI-induced PMOS transistor aging, by utilizing their predictive characteristics. Effective failure prediction requires design of effective and efficient sensors. New aging sensor designs combining the concepts of pre-sampling and stability checking enable efficient and effective failure prediction at a very low cost. System performance is actually improved because failure prediction assisted by the new aging sensor designs enable close to best-case design instead of traditional worst-case design. It is also demonstrated in this paper that simple

temperature sensors are useful but not sufficient for NBTI-induced PMOS aging.

Our future research directions include: 1. Application of the concept of circuit failure prediction for other reliability mechanisms such as infant mortality; 2. Techniques for combining circuit failure prediction with on-line periodic self-test and concurrent error detection techniques, e.g., residue codes may be used for end-to-end datapath checking at a low-cost; 3. Techniques for self-correction / self-adaptation and quantification of their benefits; and, 4. Use of failure prediction sensors for off-line test and diagnosis such as post-silicon speed-path debug and timing diagnostics for timing yield analysis [Datta 04, Franco 91, Ghosh 06, Pant 00, Uzzaman 05, Yan 03].

Acknowledgment

This research was supported in part by Focus Center Research Program Center for Circuit and System Solutions (www.c2s2.org). We thank Eugene Kishinevsky of Stanford University for his help with the synthesis of the OpenRISC processor core, and Kee Sup Kim, Jose Maiz and James Tschanz of Intel for helpful comments.

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